



Multi-Sensor-Based Long-Range Precision Landing on a High-Speed Mobile Drone Station

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Abstract

Battery-powered multicopter drones are constrained by limited range and endurance due to the low energy density of onboard batteries. To overcome these limitations, mobile drone stations capable of battery charging and replacement have been introduced. However, achieving precision landings on high-speed mobile stations—especially when they are initially beyond the visual range of the drone’s onboard sensors—remains a significant challenge. To address this, this paper proposes a precision landing approach that enables the drone to reliably land on a high-speed mobile drone station, even when the station initially lies beyond the visual range of the onboard camera. Specifically, we present a multi-sensor-based localization method that leverages the complementary characteristics of the global positioning system, ultra-wideband, and camera vision’s sensory data to robustly estimate the relative position and orientation of the mobile station. We further introduce a dynamically feasible three-phase flight guidance strategy that aligns the drone’s motion with the station’s velocity, ensuring smooth and accurate landings. Building upon this, to support the proposed method, we design a dedicated hardware configuration for both the drone and the mobile station. We validate our approach through real-world flight experiments, demonstrating consistent and successful landings on a mobile drone station traveling at speeds up to 25 km/h and starting from distances exceeding 20 m.

Keywords Mobile drone station · Multi-sensor-based localization · Precision landing · Ultra-wideband

1 Introduction

Drones are increasingly being deployed across a wide range of applications and mission scenarios. Among the various types of drones, battery-powered multi-copter drones are the most commonly used, owing to their low production cost,

simple mechanical structure, and ease of control. However, limited battery capacity inherently constrains their operational range and mission duration. One promising solution to address this limitation is the use of a mobile drone station. A mobile drone station is a modular system that houses a drone, provides a landing pad for safe takeoff and landing, and includes automated battery charging or replacement functionality [1]. By providing automated battery charging or replacement during the mission, the system mitigates the limitations imposed by the drone’s battery capacity.

The mobile drone station is typically coupled with autonomous landing capabilities to fully leverage the advantage of automated battery charging. In particular, the capability for a drone to land on a high-speed station starting from a long distance substantially reduces mission downtime related to battery charging and replacement. The precision landing on the moving station requires technology that accurately estimates the relative position of the station with respect to the drone. GPS, primarily used in outdoor environments, exhibits errors of approximately 10 m, rendering it unsuitable for precision landing on the mobile drone station measuring around 1 m in size. Therefore, achieving accurate localization

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of the mobile drone station requires the integration of additional sensors with higher precision than GPS (e.g., UWB, Vision with fiducial markers). To address this, we propose a multi-sensor-based long-range precision landing method applicable to a high-speed mobile drone station.

Our key contributions regarding the precision landing on the mobile drone station in motion are as follows.

- (1) We propose a multi-sensor-based station localization method that adapts to the relative distance between the drone and the station.
- (2) We introduce a flight path generation and following strategy that aligns the velocity of the drone with the station's motion.
- (3) We develop a flight guidance method that is model-free for both a drone and a station, enabling the drone to land on a high-speed station from a long range.
- (4) We design hardware configurations for both the drone and the mobile drone station to enable precision landing on the high-speed mobile drone station.
- (5) We validate the proposed system and method through real-world flight experiments.

The remainder of this paper is organized as follows: Sect. 2 reviews related work on precision landing. Section 3 describes multi-sensor-based localization methods for the station. Building on this, Sect. 4 presents a flight guidance approach that enables the drone to land on a high-speed station based on the station localization. Section 5 then presents real-world flight experiments and analyzes the results. Finally, Sect. 6 concludes the paper.

2 Related Work

Precision landing is a critical capability for achieving full autonomy in drone operations. Numerous studies have actively investigated this topic [2, 3]. A notable milestone was 2017 MBZIRC competition [4–9], which included a mission requiring a drone to autonomously land on a vehicle moving at 15 km/h. [4–7] employed MPC (Model Predictive Control), which requires a dynamic model of the drone. Specifically, [4, 5] used a dynamic model of the vehicle as well, and they utilized RTK (Real-Time Kinematic)-GPS for drone localization. [8] also relied on a dynamic model for a nonlinear controller. In contrast, [9] employed a model-free visual servoing approach. Despite the diverse methods proposed, the competition was held in a constrained environment with vehicle speeds below 15 km/h and relied solely on visual detection of a fiducial marker. This restricted the applicability of the approaches to more general or high-speed scenarios.

Beyond competitions, [10–13] address autonomous drone landing on moving platforms in practical contexts. In [10],

a drone lands on a ground vehicle equipped with a Stewart platform, demonstrating capability beyond visual range. However, the system relies on RTK-GPS and has not been validated at speeds above 20 km/h. In [11], an indoor experiment combines UWB and vision-based localization, yet the speed is low, and the test environment is spatially constrained. [12] presents a vision-based landing without RTK-GPS or drone dynamics, but it only works within visual range and is validated only at low speeds. [13] proposes an MPC-based high-speed landing at 12 m/s from long range, but is validated only in simulation.

As summarized in Table 1, our approach offers several advantages over prior works. First and foremost, while prior works are constrained in speed, reliance on environments, our method enables precision landing from 20 m away at speeds over 20 km/h. Furthermore, our method operates robustly without requiring RTK-GPS. Although RTK-GPS provides high-precision positioning, its operational flexibility is constrained by high cost, the time required to acquire an initial fix, and reliance on a dedicated base station, which restricts the operational range. Additionally, we employ a model-free approach for both the drone and the station. We define model-free as an approach that does not require a dynamic model. The presence of significant modeling challenges for both the station and the drone in our scenario necessitates this model-free approach. The station's dynamics are unknown to the drone because the mobile station can be mounted on any platform, causing the model to change with each deployment. Furthermore, even if a model is identified, the station is passive and non-cooperative, making its motion fundamentally unpredictable. Although a drone's dynamics can typically be modeled, our scenario involving high-speed, low-altitude flight presents challenges in altitude estimation, which in turn leads to significant modeling inaccuracies. A model-free approach is therefore well-suited to overcome these challenges. Moreover, our approach is validated in real-world experiments.

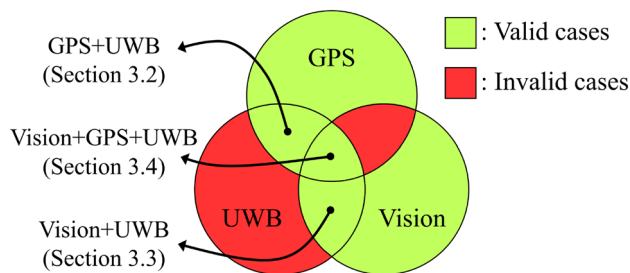
3 Multi-Sensor-Based Station Localization

We use three types of sensors (i.e., Vision, UWB, and GPS) to estimate the position of the station. The vision sensor, which is a monocular camera, detects a fiducial marker attached to the top of the station and estimates the full 6-DoF (Degree-of-Freedom) relative pose of the station, including both position and orientation. The UWB sensor measures the distance between a tag and an anchor. Unlike indoor setups with multiple anchors for 3-DoF localization [14, 15], our outdoor scenario with a single station provides only 1D range. Lastly, the GPS provides the 3-DoF global position of the station.

The available range of these sensors decreases in the order of GPS, UWB, and Vision, whereas the measurement accu-

Table 1 Comparison of related work

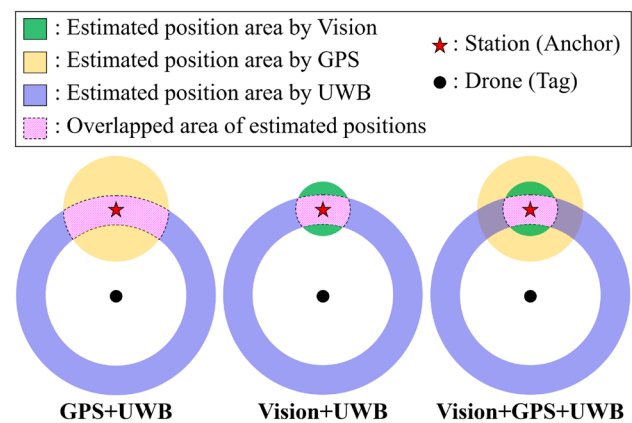
	Long-range	High-speed	w/o RTK-GPS	Real-world	Not constrained space	Drone model-free	Station model-free
[4, 5]	✗	✗	✗	✓	✗	✗	✗
[6, 7]	✗	✗	✓	✓	✗	✗	✓
[8]	✗	✗	✓	✓	✗	✗	✗
[10]	✓	✗	✗	✓	✓	✗	✓
[11]	✓	✗	✓	✓	✗	✓	✗
[9, 12]	✗	✗	✓	✓	✗	✓	✓
[13]	✓	✓	✓	✗	-	✗	✓
Ours	✓	✓	✓	✓	✓	✓	✓

**Fig. 1** Possible combinations of three sensors

racy generally increases in the reverse order. As no single sensor consistently outperforms the others across all operating conditions, a multi-sensor-based approach is essential. By combining three sensors, each sensor's strengths offset the others' limitations, enabling robust and accurate localization in dynamic environments.

Combining the three sensors yields a total of 7 possible cases, as illustrated in Fig. 1. Among them, UWB alone does not provide 3-DoF position, and while both Vision and GPS individually offer position estimates, combining them provides no additional benefit due to the superior accuracy of Vision. Therefore, we focus on the meaningful cases—shown in green—which involve either single sensors or effective combinations. For each of these, we evaluate and compare the accuracy of station localization.

Figure 2 illustrates our rationale for three sensor combinations (i.e., GPS+UWB, Vision+UWB, and Vision+GPS+UWB). Assuming that a station is located at a known height, such as on the top of a vehicle, and Gaussian estimates of Vision and GPS, UWB provides a precise range annulus. The annuli illustrated in Fig. 2 represent the estimated station's position area relative to the drone, assuming the station lies within the UWB coverage range (e.g., 20 m radius). For instance, given a distance of 10 m and a measurement error of ± 0.5 m, the estimated station's position is constrained within a 1 m thick annulus defined by an inner radius of 9.5 m and an outer radius of 10.5 m. Combining the annulus with the large GPS uncertainty circle (i.e., GPS+UWB) or the smaller

**Fig. 2** Station's estimated area from 3 valid combinations

Vision circle (i.e., Vision+UWB) theoretically reduces the estimate error. The Vision+GPS+UWB combination appears to offer little improvement over a Vision+UWB estimate. However, such an ideal analysis overlooks the complexities of real-world dynamic scenarios. For instance, UWB reliability degrades at close proximity, and vision is susceptible to motion blur and frame drops. The use of multiple sensors leverages redundancy to mitigate such failures and improve robustness. Therefore, the Vision+GPS+UWB combination remains valid. We proceed to experimentally validate the localization accuracy of these three combinations.

3.1 Coordinate frames

As illustrated in Fig. 3, we define five coordinate frames: the global inertial frame $\{G\}$, the drone body-fixed frame $\{B\}$, the onboard camera frame $\{C\}$, the station body-fixed frame $\{S\}$, and the fiducial marker frame $\{M\}$. The fiducial marker is attached to the landing pad, while the GPS and UWB modules must be installed separately from the landing pad for safe landing. Therefore, the mobile drone station comprises $\{M\}$, which is the origin of the fiducial marker, and $\{S\}$, which is the origin of both GPS and UWB. The orientation of $\{S\}$ is measured by the GPS module's compass, aligned with the

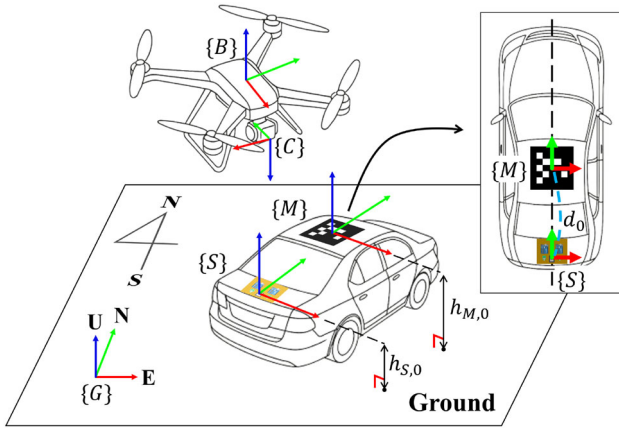


Fig. 3 Coordinate frames for our approach in motion scenario

vehicle's heading and the y-axis of $\{M\}$. Moreover, $\{M\}$ is located at a distance d_0 in the +y direction of $\{S\}$.

Precision landing requires accurate relative positioning between the drone and the station, necessitating a unified coordinate frame. Thus, all data from the four frames (i.e., $\{B\}$, $\{C\}$, $\{S\}$, and $\{M\}$) are transformed into a global ENU(East-North-Up) frame $\{G\}$, whose origin is the drone's initial GPS position. In the first step, the marker frame $\{M\}$, detected by the onboard camera, is transformed through the camera frame $\{C\}$ to the drone body-fixed frame $\{B\}$, as shown in Eq. (1).

$${}^B\mathbf{p} = {}^B\mathbf{T}_C {}^C\mathbf{T}_M {}^M\mathbf{p}, \quad (1)$$

where ${}^B\mathbf{p}$ and ${}^M\mathbf{p}$ denote points represented in frame $\{B\}$ and $\{M\}$, respectively, and ${}^B\mathbf{T}_C$, ${}^C\mathbf{T}_M$ are transformation matrices from $\{C\}$ to $\{B\}$ and from $\{M\}$ to $\{C\}$, respectively. Following this, $\{B\}$ is transformed into an ENU frame with its origin at $\{B\}$ denoted as $\{B, \text{ENU}\}$ in Eq. (2). Independently, as described in Eq. (3), the frame $\{S\}$, obtained from station's raw GPS measurement is also transformed into an ENU frame $\{B, \text{ENU}\}$.

$${}^{B, \text{ENU}}\mathbf{p} = {}^{B, \text{ENU}}\mathbf{T}_B {}^B\mathbf{p}, \quad (2)$$

$${}^{B, \text{ENU}}\mathbf{p}_S = {}^{B, \text{ENU}}\mathbf{T}_{\text{ECEF}} f_{\text{GPS} \rightarrow \text{ECEF}}({}^{\text{GPS}}\mathbf{p}_S), \quad (3)$$

where ${}^{\text{GPS}}\mathbf{p}_S$ is the frame $\{S\}$ expressed in GPS coordinates, and $f(\cdot)_{\text{GPS} \rightarrow \text{ECEF}}$ is the transformation function from the GPS frame to the Earth-Centered, Earth-Fixed (ECEF) coordinate frame.

After the transformation into the ENU frame, as described above, the position of the station is first estimated in $\{B, \text{ENU}\}$, and then transformed into $\{G\}$. However, due to large errors in GPS altitude measurements, we exclude GPS altitude from both drone and station localization. As an alternative, the height from the ground to $\{S\}$ and $\{M\}$

are assumed to be constant, and are given as known parameters $h_{S,0}$ and $h_{M,0}$, respectively (Fig. 3). This assumption implies that the vehicle moves over a flat terrain without any elevation variation. On the other hand, the drone's height is defined relative to the flat terrain and is estimated by fusing measurements from the onboard IMU (Inertial Measurement Unit) and barometer using an EKF (Extended Kalman Filter) framework.

3.2 GPS+UWB

As previously discussed, while a UWB offers more accurate distance than GPS, UWB alone does not provide 3-DoF position. To address this, we first compute the position vector of the station relative to the drone with Equation (3). The vector's magnitude is then scaled by the ratio of the UWB distance d_{UWB} to the GPS distance $\|\mathbf{r}_{\text{GPS}}\|$, denoted as λ_1 . After adding the scaled vector to the drone's position, ${}^G\mathbf{p}_B = ({}^Gx_B, {}^Gy_B, {}^Gz_B)^T$, as shown in Eq. (4), the estimated station's position in $\{G\}$ is obtained by replacing the z-component with $h_{S,0}$.

$$\left({}^G\mathbf{p}_S\right)_1 = {}^G\mathbf{p}_B + \lambda_1 {}^{B, \text{ENU}}\mathbf{p}_S, \quad (4)$$

where $\lambda_1 = \frac{d_{\text{UWB}}}{\|\mathbf{r}_{\text{GPS}}\|}$, and the GPS distance is calculated using the magnitude of a vector $\mathbf{r}_{\text{GPS}} = ({}^{B, \text{ENU}}x_S, {}^{B, \text{ENU}}y_S, -{}^Gz_B + h_{S,0})^T$.

3.3 Vision+UWB

To combine vision and UWB data, the marker position vector ${}^{B, \text{ENU}}\mathbf{p}_M$ is scaled using the ratio of the UWB distance d_{UWB} to the vision distance $\|\mathbf{r}_{\text{Vision}}\|$, similar to Sect. 3.2. According to Fig. 3, since the reference points for the two distances are different, the estimated marker origin is virtually shifted in order to align it with UWB origin at $\{S\}$. The shifted marker origin ${}^{B, \text{ENU}}\mathbf{p}_{M'}$ is calculated in frame $\{B, \text{ENU}\}$ as shown in Eq. (5).

$$\begin{aligned} {}^{B, \text{ENU}}\mathbf{p}_{M'} &= {}^{B, \text{ENU}}\mathbf{p}_M \\ &\quad + (-d_0 \sin \alpha_S, -d_0 \cos \alpha_S, -h_{M,0} + h_{S,0})^T, \end{aligned} \quad (5)$$

where α_S denotes the azimuth angle of frame $\{S\}$. In the ENU frame, the azimuth angle is defined as 0 at North and increases clockwise, satisfying $0 \leq \alpha_S < 2\pi$. Using ${}^{B, \text{ENU}}\mathbf{p}_{M'}$, the station's position is estimated in $\{G\}$, as described in Eq. (6).

$$\left({}^G\mathbf{p}_S\right)_2 = {}^G\mathbf{p}_B + \lambda_2 {}^{B, \text{ENU}}\mathbf{p}_{M'}, \quad (6)$$

where $\lambda_2 = \frac{d_{\text{UWB}}}{\|\mathbf{r}_{\text{Vision}}\|}$, and the vision distance is computed with $\mathbf{r}_{\text{Vision}} = ({}^B\text{ENU}x_{M'}, {}^B\text{ENU}y_{M'}, -{}^Gz_B + h_{S,0})^T$.

3.4 Vision+GPS+UWB

To combine all three data, ${}^B\text{ENU}\mathbf{p}_{M'}$ is applied here as well. We first compute ${}^G\mathbf{p}_{M'}$, the estimate obtained from vision, as described in Eq. (7), and then calculate the linear combination of the estimate ${}^G\mathbf{p}_{M'}$, and $({}^G\mathbf{p}_S)_1$, which combines GPS and UWB data, using the weights ω_1 and ω_2 as shown in Eq. (8).

$${}^G\mathbf{p}_{M'} = {}^G\mathbf{p}_B + {}^B\text{ENU}\mathbf{p}_{M'}, \quad (7)$$

$$({}^G\mathbf{p}_S)_3 = \omega_1 {}^G\mathbf{p}_{M'} + \omega_2 ({}^G\mathbf{p}_S)_1 \quad (8)$$

The positions estimated in Eqs. (4), (6), and (8) are all expressed with respect to the origin of frame $\{S\}$, which serves as a common reference for combining data from multiple sensors. However, the actual landing point for the drone is the center of the marker. Therefore, a coordinate transformation is applied as shown in Eq. (9) to estimate the origin of $\{M\}$ in the global frame $\{G\}$.

$${}^G\mathbf{p}_M = {}^G\mathbf{p}_S + (d_0 \sin \alpha_S, d_0 \cos \alpha_S, -h_{S,0} + h_{M,0})^T, \quad (9)$$

4 Flight Guidance for Precision Landing

The precision landing procedure is divided into three primary phases: (1) Trajectory generation and following, (2) Deceleration, and (3) Terminal descent with PID(Proportional–Integral–Derivative) control. The overall landing procedure is depicted in Fig. 4 as a state machine. The landing procedure begins by ensuring that the drone is at an initial point. Once the drone reaches the initial point, the drone generates a trajectory toward a landing point above the station and follows the trajectory. Since the station is moving, the drone must fly at a higher speed to catch up and align with the station. To achieve this, the horizontal speed of the drone is consistently set to exceed that of the station by a fixed margin v_0 .

As the drone approaches the station, it continuously evaluates the horizontal distance to the station. When the distance falls below a threshold, the drone stops trajectory generation and transitions to the deceleration phase. The objective of this phase is to reduce the relative horizontal velocity between the drone and the station to zero. Since the drone is initially faster than the station in this phase, a smooth and controlled deceleration process is crucial. To achieve this, the target speed is discretized into a sequence of decremental values. During the deceleration, the drone maintains a fixed height. Also, the drone's horizontal velocity always points toward the landing point, and its magnitude is limited by the current step in the target speed profile used by the PID controller.

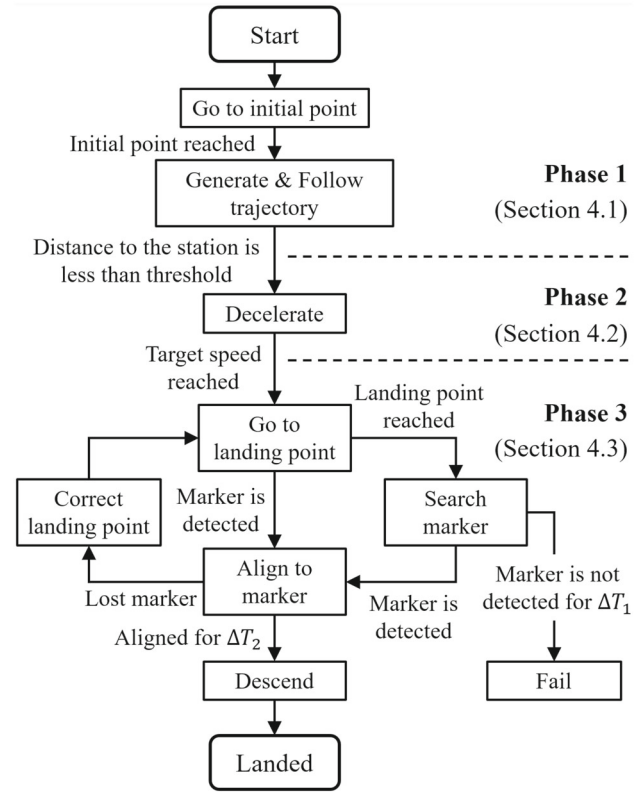


Fig. 4 State machine for the precision landing

Once the relative horizontal velocity approaches almost zero and the marker is detected, the system transitions to the terminal descent phase, which is governed by a PID controller. The horizontal velocity commands are generated based on the position error between the drone and the landing point. To ensure a safe and accurate descent, descent commands are only issued when two conditions are satisfied: (1) The horizontal distance between the drone and the marker falls below a threshold. (2) Condition (1) has been satisfied for ΔT_2 . When both conditions are satisfied, the drone begins descending at a fixed vertical speed.

4.1 Trajectory Generation and Following

4.1.1 Minimum Snap Trajectory

A minimum snap trajectory is a smooth, piecewise polynomial path optimized to minimize the integral of the square of the snap (i.e., the 4th derivative of position) while passing through specified keyframes $\sigma_T(t_i)$ in 3D space and yaw angles [16]. The minimum snap trajectory minimizes the following cost function:

$$\min \int_{t_0}^{t_m} \mu_r \left\| \frac{d^4 \mathbf{r}_T}{dt^4} \right\|^2 + \mu_\psi \left(\frac{d^2 \psi_T}{dt^2} \right)^2 dt, \quad (10)$$

$$\text{s.t. } \sigma_T(t_i) = [x_i, y_i, z_i, \psi_i]^T, \quad i = 1, \dots, m, \quad (11)$$

$$\text{continuity of } \mathbf{r}_T, \dot{\mathbf{r}}_T, \ddot{\mathbf{r}}_T, \ddot{\psi}_T, \psi_T, \dot{\psi}_T \text{ at } t_i. \quad (12)$$

where $\frac{d^4 \mathbf{r}_T}{dt^4}$ is the snap, and $\frac{d^2 \psi_T}{dt^2}$ represents angular acceleration. μ_r and μ_ψ are constants that make the integrand nondimensional. Here, t_0 and t_m are the start and end times of the trajectory, respectively, and $T = t_m - t_0$.

To safely land on a high-speed station, achieving zero relative velocity is essential. The drone generates a minimum snap trajectory to guide it along a linear path aligned with the station's velocity. This trajectory generation process requires two inputs derived from the station's GPS data, which is transmitted to the drone via a communication link. The first is the station's velocity, taken directly from the GPS data, and the second is the station's position, estimated from the same data using the method described in Sect. 3. The trajectory is defined by three keyframes: start, intermediate, and endpoint. The start point is the drone's position for the initial path generation. Thereafter, to ensure a smooth connection, the start point is updated to the last target point the drone was following on the previously generated path. The end point, identical to the landing point, is located at a height h_0 above the station marker. To quickly align the drone with the path, the intermediate point is set on the 2D xy -plane of $\{G\}$ by first projecting the trajectory start point perpendicularly onto a line parallel to the station's velocity through the end point. The intermediate point divides the segment from this foot to the end point in a 1:2 ratio. Real-time generation and tracking of this trajectory enables the drone to achieve zero relative velocity, even when the station moves along a curved path.

4.1.2 Pure Pursuit

To follow the generated trajectory, we adopt the computationally efficient Pure Pursuit algorithm [17]. By using only the geometric properties of the minimum snap trajectory, this model-free method is robust to modeling errors and avoids the computational load of the controller. This allows the drone to dynamically adjust its speed to the station's motion through a velocity control scheme. The horizontal velocity of the drone, which is managed by the PX4's internal PID controller, is directed toward a target point on the path and set to exceed the station's speed by v_0 , while a PD controller independently adjusts vertical velocity and yaw to match the target point and heading. At each control loop, the horizontal distance to the target point is measured, and when it falls below a look-ahead distance, the target point is updated to the next point. The look-ahead distance is set based on the

drone's speed, with a minimum value to ensure both tracking accuracy and smoothness.

4.2 Deceleration

As the drone approaches the station, the drone follows the station from behind, aligned with the station's velocity. Consequently, the path becomes almost linear, making additional path generation unnecessary. Therefore, once the horizontal distance between the drone and the station d falls below a threshold l , the drone stops generating the path and switches the reference point to the landing point. The drone then initiates deceleration to match the station's speed, employing PID control and a discretized stepwise speed limiting strategy. The stepwise speed limits are determined by dividing the velocity difference between the drone and the station into n discrete steps at the start of the deceleration phase, where each step is defined according to the current horizontal distance d , as follows:

$$v_{\text{limit}} = v_{\text{drone}} - \left(\frac{v_{\text{drone}} - v_{\text{station}}}{n} \right) i, \quad (13)$$

$$i \in \{1, \dots, n\} \text{ such that } \frac{l}{n}(i-1) \leq d < \frac{l}{n}i, \quad (14)$$

where v_{limit} is the drone's speed limit, v_{drone} is the drone's speed, and v_{station} is the station's speed.

4.3 Terminal descent with PID control

During deceleration, if the drone detects the marker or reaches the landing point, the speed limit is lifted and the system transitions to terminal descent. If the marker is not detected due to errors in station localization, the drone initiates a search by ascending to a height h_{max} . After reaching h_{max} , if the marker remains undetected for ΔT_1 , the landing is aborted and the drone hovers.

Once the marker is detected, the drone maintains its height and aligns horizontally and in yaw with the marker using separate PID controllers. While the marker is detected, the system records the horizontal offset e_{Vision} between the GPS- and marker-based positions. The recorded e_{Vision} corrects the landing point when the marker is lost. After the marker is detected again, the horizontal distance error stays within a threshold for ΔT_2 , the drone begins descending at a constant vertical speed. If the marker becomes undetectable due to the limited onboard camera's FOV(Field of view) near the marker, the drone continues descending at a horizontal velocity matched to that of the station.

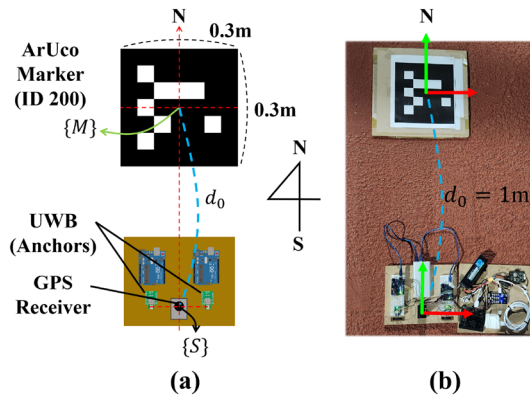


Fig. 5 Configuration of the mobile drone station: (a) design of the station, (b) experimental setup

5 Experiments and Results

5.1 Multi-Sensor-Based Station Localization

We use a Holybro-F9P GPS module, a DWM1000 UWB module with a range of up to 20m, and a camera with a 0.3×0.3 m ArUco marker (id=200). The camera has a resolution of 640×480 , a 100° FOV, and a framerate of 30 fps. The drone receives the station's GPS data, including position and velocity, via a Wi-Fi link, actively measures its distance to the station anchor using UWB, and passively detects the ArUco marker. Using the same three sensors, we configure both the station and the drone. The station setup includes two UWB anchors, whose average measurement is used for robustness (Fig. 5-a). The drone configuration includes a top-mounted UWB tag, a GPS module, and a camera mounted in a downward-facing orientation relative to the body-fixed frame (Fig. 6-a). While an obliquely mounted camera allows for earlier marker detection during the approach phase, it inevitably loses sight of the marker during the terminal descent phase. In contrast, the downward-facing camera guarantees that the marker is consistently visible during the terminal descent phase of the landing. The drone also mounts four electromagnets on its landing gears, as part of the system shown in Fig. 6-b, to enable safe landing on a high-speed station. The drone's onboard computer, *NVIDIA Jetson Orin NX*, controls the electromagnetic system, and we use the PX4 for the flight controller.

To evaluate the localization accuracy of the station using three sensors, we deploy the station in a stationary position on the ground as illustrated in Fig. 5-b, where the y-axes of $\{S\}$ and $\{M\}$ are aligned with magnetic North for convenience, and the origin of $\{S\}$ is measured using RTK-GPS with $d_0 = 1$ m. The drone's initial position, which defines the origin of the ENU frame $\{G\}$, is also measured using RTK-GPS. We then conduct flight experiments, during which the drone maintains a height of 2 m and collects sensor data for

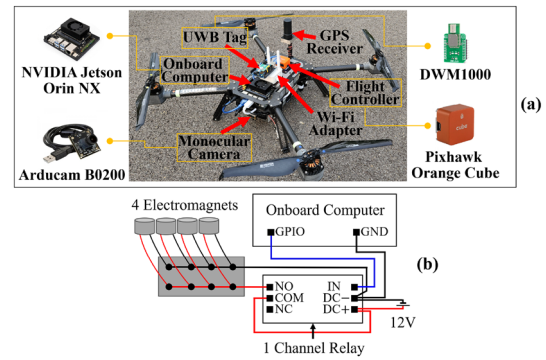


Fig. 6 Configuration of (a) the drone and (b) the electromagnetic system

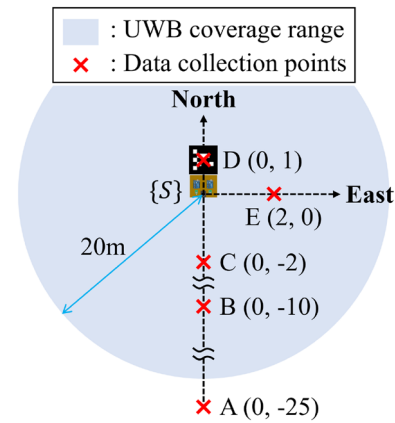


Fig. 7 Data collection points for station localization

approximately 20 s at each of the four locations (i.e., A, B, C, and D) along the y-axis of $\{S\}$, and also at E(2,0) to simulate marker loss as shown in the Fig. 7. The locations are selected based on the available range of the sensors. A represents the area outside the UWB coverage range, B is the area within the UWB coverage range with the highest reliability, C is where UWB reliability decreases, and D represents the area where marker detection is possible. At each point, the origin of $\{M\}$ is estimated based on the available sensors as described in Sect. 3.

The reliability of UWB measurements is determined based on three exceptional conditions: (1) At least one of the two anchor measurements is missing, (2) The UWB distance differs by more than 3 m from the previous measurement, (3) the measurement falls outside the valid range ($0 \leq d_{UWB} \leq 20$ m). Due to the characteristics of UWB, these exceptional cases occur more frequently when the tag is located very close to the anchors [18].

The station localization results are presented in Table 2 and Fig. 8. The Vision, UWB, and GPS operate at 30 Hz, 20 Hz, and 10 Hz, respectively, and the frequency of the combined output is determined by the fastest sensor. That is, our approach continuously updates the station's location using

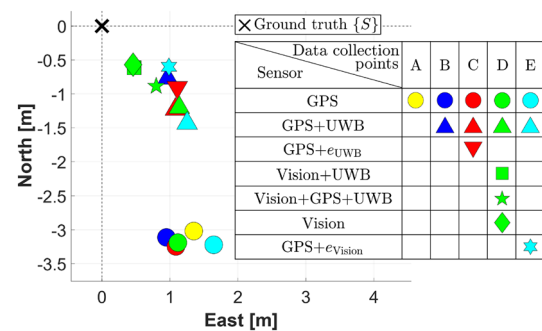
Table 2 Comparison of station localization accuracy

Data collection points in {S}	Sensor	Average x-error [m]	Average y-error [m]	RMSE [m]	Standard deviation [m]	Frequency [Hz]	UWB Reliability
A	GPS	1.35	-3.02	3.31	0.010	10	-
B	GPS	0.95	-3.11	3.25	0.013	10	-
	GPS+UWB	0.97	-0.77	1.24	0.466	20	0.90
C	GPS	1.09	-3.24	3.42	0.048	10	-
	GPS+UWB	1.08	-1.23	1.64	0.047	20	0.27
	GPS+ e_{UWB}	1.11	-0.90	1.43	0.048	20	-
D	GPS	1.12	-3.19	3.38	0.020	10	-
	GPS+UWB	1.14	-1.20	1.65	0.062	20	0.15
	Vision+GPS+UWB	0.80	-0.89	1.19	0.032	30	0.15
	Vision+UWB	0.47	-0.61	0.78	0.070	30	0.15
	Kalman filter	0.45	-0.60	0.75	0.033	30	0.15
	Vision	0.46	-0.57	0.73	0.021	30	-
E	GPS	1.64	-3.22	3.62	0.019	10	-
	GPS+UWB	1.25	-1.43	1.90	0.240	20	0.73
	GPS+ e_{Vision}	0.90	-0.60	1.15	0.019	20	-

Best results are given in bold

the fastest sensor, while incorporating the most recent data from slower ones. This produces the smooth, high-frequency localization crucial for real-time, high-speed landing maneuvers. At points B, C, D, and E, where UWB is available, the RMSE of GPS alone is 3.42 m, while GPS+UWB achieves an average RMSE of 1.62 m, corresponding to a 52.6% reduction in error. This demonstrates that GPS+UWB provides significantly improved accuracy over GPS alone. At point D, where the vision sensor is available, the use of the vision sensor consistently results in better accuracy than GPS+UWB. To evaluate the performance of our proposed model-free methods, we also conduct a comparative analysis against a model-based Kalman filter. The Kalman filter baseline assumes a 2D constant velocity model for the station's dynamics. The filter fuses measurements from three different sensors, updating asynchronously according to each sensor's frequency. The resulting state estimate is published at 30 Hz. Among the methods evaluated, the vision-only approach yields the best performance with an RMSE of 0.73 m, which is superior to an equally weighted combination of all sensors, as formulated in Eq. 8. Furthermore, this method demonstrates superior performance compared to the Kalman filter, even in an ideal scenario where the station adheres to a constant velocity model. This result indicates that the best combination in Eq. 8 is to set the weight ω_1 to 1 and ω_2 to 0. This vision alone error includes the drone's GPS error.

In Table 2, the UWB reliability is lower at points C, D, and E, compared to B. Notably, point D, where the ArUco marker is detected, shows the lowest reliability of 0.15. Although points C and E have the same distance, E shows approxi-

**Fig. 8** Station localization errors under distinct sensor suite-based methods

mately 2.7 times higher reliability than C. This suggests that the reduced reliability at close range is not solely due to proximity, but also due to the Line-of-Sight blockage caused by the drone's body. To address the issue of reduced UWB reliability at close distances, we record the average offset e_{UWB} between the GPS-only and GPS+UWB estimates at B, which has the highest reliability. When the drone reaches C, this recorded offset e_{UWB} is added to the GPS-only estimate, resulting in 58% reduction in localization error. Similarly, to assist recovery after the marker is lost, we record e_{Vision} at D. At point E, adding the recorded e_{Vision} to the GPS-only estimate results in a 68% error reduction, improving localization accuracy even in the absence of marker detection.

Based on the accuracy evaluation of the station localization, we adopt the multi-sensor-based strategy described in Eq. (15). When UWB data is unavailable, the station's position is estimated using GPS only. Once UWB becomes

available, we use the GPS+UWB method. Furthermore, if the drone approaches within 4 m of the station, e_{UWB} is added to the GPS-only estimate to improve accuracy. When the drone is close enough that the marker is detected, localization is performed using vision-only estimates. During this phase, e_{Vision} is also recorded. If the marker is lost, the station's position is estimated by adding the recorded offset e_{Vision} to the GPS-only estimate.

Multi-sensor-based method =

$$\left\{ \begin{array}{ll} \text{GPS} & \text{if } d > 20 \text{ m,} \\ \text{GPS+UWB} & \text{if } 4 \text{ m} < d \leq 20 \text{ m,} \\ \text{GPS} + e_{\text{UWB}} & \text{if } 0 < d \leq 4 \text{ m and} \\ & \text{Marker never detected,} \\ \text{Vision} & \text{if } 0 \leq d \leq 4 \text{ m and} \\ & \text{Marker is being detected,} \\ \text{GPS} + e_{\text{Vision}} & \text{if } 0 \leq d \leq 4 \text{ m and} \\ & \text{Marker was detected before,} \end{array} \right. \quad (15)$$

5.2 Precision Landing on a High-Speed Mobile Drone Station

To validate the proposed precision landing approach, we conduct flight experiments with key parameters summarized in Table 3. Among the parameters, v_0 must be set according to the drone's performance limits, while the optimal value for ΔT_2 should be found by starting at 5 s and gradually decreasing it until the phase transition occurs. Additionally, to ensure a smooth deceleration, the transition distance l is set to be proportional to the relative velocity v_0 , with more distance added at higher speeds. For the experiment, a car, manually operated by a human driver on public roads, serves as the mobile platform for carrying the mobile drone station. Moreover, our implementation of minimum snap trajectory generation is based on the open-source code provided in [19].

The results at 20 km/h station speed are shown in Fig. 9 and Fig. 10. The drone initiates the landing from a position where the horizontal distance to the station is 28.2 m. As seen in Fig. 9-a, the drone follows a minimum snap trajectory, which is generated at 1 Hz, using a pure pursuit that operates with a 20 Hz loop frequency in Phase 1. At the beginning, the limited sensor range leads to GPS-only station estimates, causing oscillations (Fig. 9-b). However, once the drone enters the UWB coverage range, the position estimation of the station stabilizes. During the final descent, the marker becomes undetectable due to the camera's FOV limitation, causing the update of the station's position estimate to halt. Following the last detection of the marker, the drone continues its descent and completely lands at a location approximately 6.3 m horizontally away from the last detected marker position.

Table 3 The user-defined parameters for experiments

Symbol	Description	Value
$h_{M,0}$	Height of the frame $\{M\}$	1.5 m
$h_{S,0}$	Height of the frame $\{S\}$	1.0 m
d_0	y-offset between frames $\{M\}$ and $\{S\}$	1.3 m
h_0	Height offset of landing point from $h_{M,0}$	2.2 m
$h_{\max}$	Max height from $h_{M,0}$ for search	3.0 m
v_0	Speed margin	2.0 m/s
l	Transition distance from Phase 1 to 2	$3v_0 + v_{\text{drone}}$
n	# of discrete steps for deceleration	6
ΔT_2	Descent start duration	3 Sec

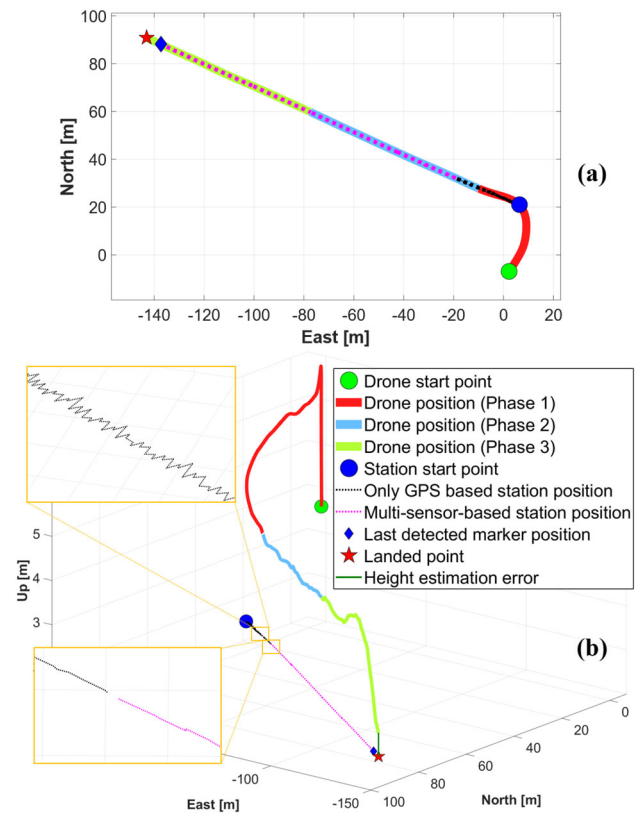


Fig. 9 Trajectory of the drone and the station during the precision landing: **a** xy-plane 2D plot, **b** 3D plot

Figure 10 illustrates the drone velocity commands and the velocities of both the station and the drone during the landing procedure. Figure 10-a shows the generated horizontal velocity commands to the drone, as well as the horizontal speeds of both the drone and the station until the final touchdown. Additionally, it presents the horizontal distance between the drone and the station estimated according to Equation (15). During Phase 1, the horizontal velocity command to the drone is generated adaptively to exceed the station's speed by a constant margin v_0 . Hence, the horizontal speed gap is fixed at v_0 throughout Phase 1. When the horizontal distance between

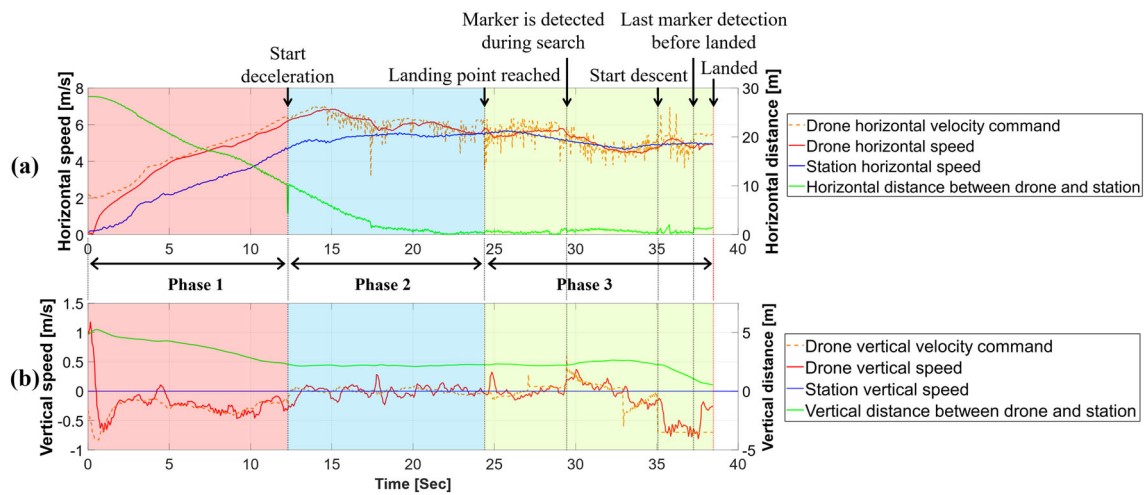


Fig. 10 Time-series comparison of drone velocity commands, response speed, station speed, and relative distance: **a** horizontal **b** vertical

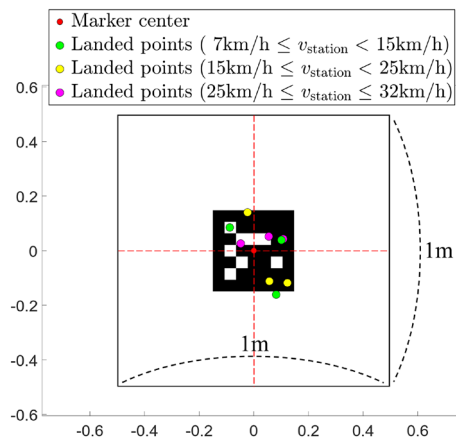


Fig. 11 Landed points under varying station speeds

the drone and the station reaches approximately 10 m, the system transitions to Phase 2, during which the drone begins to decelerate. From Phase 2, the horizontal velocity command is computed via PID control at 20Hz, with the stepwise speed limit gradually reduced for smooth deceleration. As PID control begins, the command exhibits oscillations. However, due to the imposed speed limit, the variations remain within a constrained range, and as a result, the relative speed between the drone and the station progressively diminishes, approaching zero at the landing point. In Phase 3, similar command oscillation is observed, but within acceptable limits, satisfying descent conditions. As the descent progresses and the marker is intermittently lost, sharp changes in the estimated position result in increased fluctuations in the velocity command. After the final marker detection, a smooth velocity command is generated using a weight of 1.1 relative to the station's velocity to compensate for communication and control delays.

Table 4 Comparison of precision landing errors over five trials for each station speed intervals

Station speed intervals [km/h]	Average error of the best three landings [m]	Success rate [%]
1) $7 \leq v_{\text{station}} < 15$	0.137	100
2) $15 \leq v_{\text{station}} < 25$	0.146	80
3) $25 \leq v_{\text{station}} \leq 32$	0.082	60

Figure 10-b shows the generated vertical velocity commands to the drone. The commands are computed based on PD control using the vertical displacement of the target height with respect to the drone. This excludes the terminal constant descent command, which is issued after the descent condition is satisfied in Phase 3. Throughout the entire process, the vertical velocity is limited to a maximum of 0.7 m/s. In Phase 1, the command supports trajectory following. As the drone enters Phase 2 and reaches the height of the landing point, the vertical velocity command approaches zero to maintain this height. When the drone reaches the landing point, the system transitions to Phase 3. In Phase 3, due to the marker not being detected, an ascent command is issued for search. During the search, once the marker is detected, the drone ascends further to ensure stable marker detection and then aligns with the marker upon reaching the target height. If the descent conditions are satisfied thereafter, the drone descends with a constant vertical velocity command of -0.7 m/s until the landing is complete. When the drone completes landing, the drone's height estimation error results in a distance gap of approximately 0.5 m.

As additional experiments, we conduct five repeated trials for each of the three station speed intervals to further vali-

date the proposed precision landing method. The intervals are defined based on threshold speeds of 15 km/h and 25 km/h, with the station speed ranging from a minimum of 7 km/h to a maximum of 32 km/h. These speeds represent the average speed of the station during Phase 3. The experimental results are summarized in Fig. 11 and Table 4. Figure 11 shows the best three landed points for each speed interval on a 1×1 m landing pad, while Table 4 presents the average errors of the three best trials and the success rate over five trials per speed interval.

In the trials, all landings succeed in the interval (1). In the interval (2), four out of five trials are successful, while in the interval (3), three trials succeed. The failure in (2) is due to a timeout in marker detection caused by a station localization error, while the failure in the (3) results from the drone attempting to land at the front or rear edge of the landing pad. Although the x -axis error from the marker is within the landing pad, landing at the front or rear edge is not due to localization error but because the speeds of the station and the drone are not matched during the terminal descent. A comparison of the errors in Table 4 shows that a lower station speed does not necessarily lead to a smaller error. This is because the landing error is primarily influenced by the relative velocity between the station and the drone at the moment of landing, rather than the station's absolute speed. Additionally, we analyze the result as a combined effect of the drone's velocity control PID gain values and the parameters listed in Table 3. The overall average error is 0.121 m, approximately half the width of the landing gear (0.25 m) of the drone used in the experiments, indicating an acceptable level of error.

We conduct an additional validation to further evaluate the generalization performance of our proposed method, which is detailed in Sect. 3 and Sect. 4. Using the Gazebo simulator, we confirm that precision landing is possible even when the station moves along a curved path. The station moves along the curved path, maintaining a speed of 14.4 km/h and rotating at 86 deg/s around the station's body-fixed vertical axis. As shown in Fig. 12-a, the drone progressively tracks the station along the station's curved path. The drone achieves this tracking by updating its reference trajectory in real-time based on the station's position, estimated by Equation (15), and its velocity, following the path generation method described in Sect. 4. In this experiment, the drone detects the marker before satisfying the landing point arrival condition, causing the drone to transition from Phase 2 to Phase 3 in accordance with the state machine shown in Fig. 4. Consequently, in Phase 3 (as shown in Fig. 12-b), the drone does not ascend to search for the marker. Instead, the drone maintains height until the descent condition is satisfied, after which the drone completes the landing. This result confirms that the proposed method enables precision landing using only the

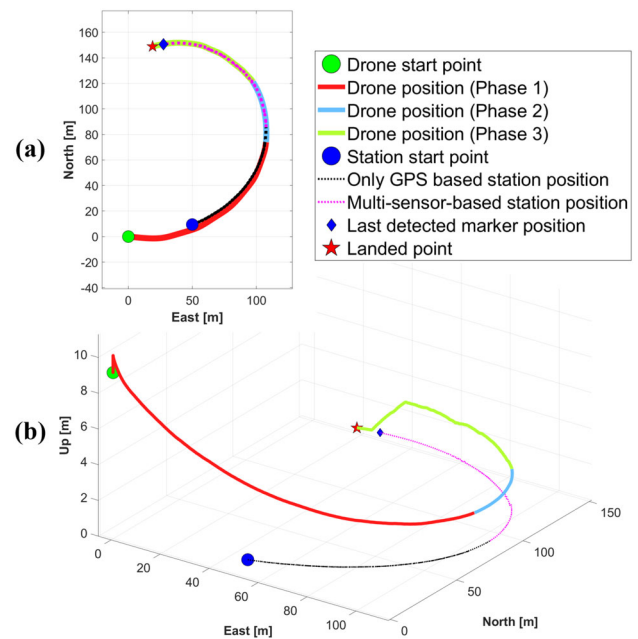


Fig. 12 Trajectory of the drone and the station during the precision landing with a curved station path in a simulation environment: (a) xy -plane 2D plot, (b) 3D plot

station's information obtained via communication, without requiring a station model or motion prediction.

6 Conclusion

We propose a long-range precision landing on a high-speed mobile drone station by combining three sensors: Vision, UWB, and GPS. We first conduct experiments to compare the localization accuracy of the station under various combinations of these sensors. The results of station localization show that combining UWB and GPS yields more accurate estimates than using GPS alone. Moreover, when the vision sensor detects a fiducial marker, using vision alone provides the highest accuracy. Accordingly, the proposed system selects the appropriate sensors for station localization based on the relative distance between the drone and the station.

The precision landing, based on the results of multi-sensor-based estimation, consists of three phases. The first phase guides the drone toward the station along a straight path aligned with the station's velocity, using a minimum snap trajectory and pure pursuit-based following. The second phase applies PID control with stepwise speed limits to decelerate the drone. The third phase performs the terminal descent with PID control. Without requiring any dynamic model of both the drone and the station, the proposed method adaptively generates velocity commands according to the changing station velocity. As a result, the proposed method achieves long-range precision landing on the high-speed mobile drone

station with an average landing error of 0.121 m at speeds up to 25 km/h.

Author Contributions Joohyuk Lee: Multi-sensor-based station localization, flight guidance, configurations, experiments, and manuscript. Hyeonjeong Heo: UWB+GPS method, UWB & fiducial marker configuration, and experiments. Hojun Lee: UWB module setup and experiments. Kyuman Lee: Supervision, manuscript review, and editing.

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Data Availability The source code developed in this study is available from the corresponding author upon reasonable request. The code is not publicly available due to restrictions regarding intellectual property and ongoing patent applications.

Declarations

Conflict of interest All authors declare that there is no competing financial interest or personal relationship that could have appeared to influence the work reported in this paper.

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